APPLICATION USER GUIDE & WORKOUT MANUAL

Early Disease Risk Prediction Using Explainable Machine Learning (XGBoost, SHAP, MySQL & Ollama)

MANDATORY MEDICAL & EDUCATIONAL DISCLAIMER:
This software application is built strictly for educational exploration and academic research. It does NOT diagnose diseases, prescribe medical treatments, or substitute for consultation with licensed healthcare practitioners. Probability values and SHAP feature attributions describe mathematical patterns within historical research cohorts, not clinical diagnostic certainties.

1. System Overview & Technology Stack

The platform pairs predictive gradient-boosted decision trees (**XGBoost**) with cooperative game-theoretic interpretability (**SHAP TreeExplainer**) and local conversational intelligence (**Ollama with llama3.2:latest**). Data persistence is managed via **MySQL**, and the responsive user interface is powered by **React 19, Tailwind CSS v4, and Recharts**.

Component	Technology	Operational Role & Port
Frontend UI	React 19, Vite, Tailwind CSS v4, Recharts, Lucide	Responsive web dashboard (Desktop & Mobile Drawer) at **http://localhost:5173**
Backend API	FastAPI, Uvicorn, Pydantic v2	REST API endpoints, schema validation, PDF generation at **http://localhost:8000**
Database	MySQL Server (Port 3306), SQLAlchemy, PyMySQL	Persistent database **disease_risk_db** storing assessment logs & SHAP vectors
Machine Learning	XGBoost, Scikit-learn, Pandas, NumPy, Joblib	Trained classifiers for Diabetes, Heart Disease, and Kidney Disease
Explainable AI	SHAP (SHapley Additive exPlanations)	Calculates local feature shifts, base expectations, and global rankings
Local LLM	Ollama (llama3.2:latest) on http://localhost:11434	Plain-language clinical risk Q&A; with rule-based fallback if offline

2. How to Work the Application (Step-by-Step)

Step 1: Start Backend Services: Launch the MySQL database server. Open PowerShell and run:
cd backend; python -m uvicorn app.main:app --reload --port 8000
Step 2: Start Frontend Application: In a second terminal window, run:
cd frontend; npm run dev (Open browser at http://localhost:5173)
Step 3: Verify Local LLM (Optional): Ensure Ollama is active with ollama list showing llama3.2:latest.
Step 4: Select a Disease Module: Navigate via the sidebar to Diabetes, Heart Disease, or Kidney Disease.
Step 5: Enter or Load Parameters: Fill in clinical measurements or click the quick demo buttons to auto-populate samples.
Step 6: Review & Confirm: A confirmation modal displays verified physiological numbers before running inference.
Step 7: Analyze Results & SHAP Waterfall: Review the probability, illustrative risk band, and red/green feature shifts.
Step 8: Ask Ollama or Export PDF: Click 'Consult Ollama' to ask follow-up questions, or download the PDF Report.

3. Understanding Explainable AI (SHAP)

SHAP values measure the positive or negative contribution of each clinical parameter relative to the model's baseline expectation:

• **Base Value (E[f(x)]):** The average probability across the research training cohort (e.g. ~35%).

• **Positive SHAP (Red Bars):** Parameters that shifted the estimated probability *higher* (e.g. elevated plasma glucose or older age).

• **Negative SHAP (Green Bars):** Parameters that acted as mitigating or protective factors, shifting the score *lower* (e.g. normal serum creatinine).

• **Important Rule:** High SHAP magnitude proves that the decision tree utilized that feature for leaf splits; it does not prove clinical causality.

4. Concrete Workout Examples (Hands-On Testing Guide)

Use the exact clinical profiles below to test and evaluate the system across Lower Risk vs. Elevated Risk scenarios.

Workout 1: Diabetes Risk Assessment (Pima Indians Cohort)

Parameter	Unit / Range	Sample A (Lower Risk)	Sample B (Elevated Risk)
Number of Pregnancies	0 - 25 count	1	4
Plasma Glucose (2-hr OGTT)	40 - 450 mg/dL	94	168
Diastolic Blood Pressure	30 - 200 mm Hg	68	86
Triceps Skinfold Thickness	5 - 110 mm	20	34
2-Hour Serum Insulin	0 - 1200 µU/mL	65	195
Body Mass Index (BMI)	12.0 - 75.0 kg/m²	22.6	36.2
Diabetes Pedigree Score	0.01 - 3.00 score	0.24	0.78
Patient Age	18 - 115 years	28	54
Expected Model Output	**Illustrative Band**	**10% - 15% (Lower Risk)**	**75% - 85% (Higher Risk)**
Key Expected SHAP Drivers	**Marginal Influence**	Normal glucose & BMI push risk DOWN	Elevated Glucose, BMI & Age push risk UP

Workout 2: Heart Disease Risk Assessment (UCI Cleveland Cohort)

Parameter	Normal / Lower Risk Profile	High Risk Cardiovascular Profile
Age / Sex	44 years / Female (0)	65 years / Male (1)
Chest Pain Type (cp)	2 (Atypical Angina)	4 (Asymptomatic - High Silent Ischemia)
Resting BP / Cholesterol	118 mm Hg / 190 mg/dL	158 mm Hg / 285 mg/dL
Fasting Blood Sugar > 120	0 (False)	1 (True)
Resting ECG / Max HR (thalach)	0 (Normal) / 168 bpm	2 (LV Hypertrophy) / 122 bpm
Ex. Angina / Oldpeak ST Depression	0 (No) / 0.2 mm	1 (Yes) / 2.8 mm depression
Slope / Major Vessels (ca)	1 (Upsloping) / 0 vessels	2 (Flat) / 2 vessels colored
Thallium Stress Scintigraphy	3.0 (Normal)	7.0 (Reversible Defect)
Expected Output & Category	**12% - 20% (Lower Estimated Risk)**	**85% - 95% (Higher Estimated Risk)**
Key SHAP Drivers	Absence of major vessel disease & normal thal	ST depression, fluoroscopy vessels & thal defect

Workout 3: Chronic Kidney Disease Assessment (UCI CKD Cohort)

Diagnostic Domain	Healthy Normal Profile	Impaired Renal Profile (Elevated CKD Risk)
Vitals (Age / BP)	`Age: 35 yrs, BP: 70 mm Hg`	`Age: 62 yrs, BP: 90 mm Hg`
Urinalysis (sg, al, su)	`Specific Gravity: 1.025, Albumin: 0, Sugar: 0`	`Specific Gravity: 1.010, Albumin: 3, Sugar: 1`
Urine Cells (rbc, pc, pcc, ba)	`RBC: Normal, PC: Normal, Clumps/Bacteria: None`	`RBC: Abnormal, PC: Abnormal, Clumps: Present`
Blood Chemistry (bgr, bu, sc)	`Glucose: 92, Urea: 24, Creatinine: 0.8 mg/dL`	`Glucose: 160, Urea: 68, Creatinine: 3.2 mg/dL`
Electrolytes (sod, pot)	`Sodium: 142 mEq/L, Potassium: 4.2 mEq/L`	`Sodium: 130 mEq/L, Potassium: 5.2 mEq/L`
Hematology (hemo, pcv, wc, rc)	`Hemo: 15.6 g/dL, PCV: 46%, WBC: 6800, RBC: 5.3`	`Hemo: 9.4 g/dL, PCV: 29%, WBC: 10200, RBC: 3.6`
Comorbidities & Signs	`HTN: No, DM: No, Appetite: Good, Edema/Anemia: No`	`HTN: Yes, DM: Yes, Appetite: Poor, Edema: Yes, Anemia: Yes`
Expected Output & Category	**3% - 8% (Lower Estimated Risk)**	**92% - 98% (Higher Estimated Risk)**
Key SHAP Drivers	Normal serum creatinine & high hemoglobin	Elevated Serum Creatinine, Albuminuria & Anemia

5. Interactive Local LLM Assistant Workout

After generating any prediction, navigate to **LLM Explanation Assistant** to conduct conversational Q&A; with llama3.2:latest. Test the assistant with the following targeted prompt templates:

Inquiry Category	Sample Question to Ask Ollama Assistant	Expected Explanatory Behavior
Risk Factor Inquiry	*"Why is my plasma glucose / serum creatinine the top risk factor?"*	Directly references the observed level and SHAP contribution (+0.45), explaining how it shifts odds in decision trees without claiming causality.
Probability Meaning	*"What does an estimated probability of 78% actually mean?"*	Explains statistical likelihood in research cohorts, differentiating laboratory calibration from individual clinical diagnosis.
SHAP Directionality	*"What is the difference between red and green bars on the chart?"*	Clarifies that red represents risk-escalating features while green reflects protective or mitigating physiological measurements.
Model Boundaries	*"Can this model tell me if I definitely need medication?"*	Firmly demurs, highlighting safety boundaries, absence of clinical licensing, and mandating consultation with a licensed physician.

SUMMARY CHECKLIST FOR DEMONSTRATIONS & ACADEMIC AUDITS:

1. Confirm MySQL service is running on port 3306 with your configured credentials.
2. Confirm FastAPI backend is online at http://localhost:8000/api/health.
3. Confirm Vite frontend is responsive on both Desktop viewports and Mobile drawers.
4. Confirm that every prediction saves to MySQL and produces a downloadable ReportLab PDF.
5. Confirm that Ollama generates contextual explanations or switches gracefully to rule synthesis.